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compact1, compact2, compact3

java.util

Class IntSummaryStatistics

java.lang.Object
    java.util.IntSummaryStatistics

All Implemented Interfaces:

IntConsumer

```
public class IntSummaryStatistics
extends Object
implements IntConsumer
```

A state object for collecting statistics such as count, min, max, sum, and average.

This class is designed to work with (though does not require) streams. For example, you can compute summary statistics on a stream of ints with:

```
IntSummaryStatistics stats = intStream.collect(IntSummaryStatistics::new,
                                               IntSummaryStatistics::accept,
                                               IntSummaryStatistics::combine);
```

IntSummaryStatistics can be used as a reduction target for a stream. For example:

```
IntSummaryStatistics stats = people.stream()
                                   .collect(Collectors.summarizingInt(Person::getD
```

This computes, in a single pass, the count of people, as well as the minimum, maximum, sum, and average of their number of dependents.

Implementation Note:

This implementation is not thread safe. However, it is safe to use `Collectors.toIntStatistics()` on a parallel stream, because the parallel implementation of `Stream.collect()` provides the necessary partitioning, isolation, and merging of results for safe and efficient parallel execution.

This implementation does not check for overflow of the sum.

Since:

1.8

Constructor Summary

Constructors

Constructor	Description
`IntSummaryStatistics()`	Construct an empty instance with zero count, zero sum, `Integer.MAX_VALUE` min, `Integer.MIN_VALUE` max and zero average.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
`void`	`accept(int value)`	Records a new value into the summary information
`void`	`combine(IntSummaryStatistics other)`	Combines the state of another `IntSummaryStatistics` into this one.
`double`	`getAverage()`	Returns the arithmetic mean of values recorded, or zero if no values have been recorded.
`long`	`getCount()`	Returns the count of values recorded.
`int`	`getMax()`	Returns the maximum value recorded, or `Integer.MIN_VALUE` if

		no values have been recorded.
int	**getMin**()	Returns the minimum value recorded, or Integer.MAX_VALUE if no values have been recorded.
long	**getSum**()	Returns the sum of values recorded, or zero if no values have been recorded.
String	**toString**()	Returns a string representation of the object.

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, wait, wait, wait

Methods inherited from interface java.util.function.IntConsumer

andThen

Constructor Detail

IntSummaryStatistics

```
public IntSummaryStatistics()
```

Construct an empty instance with zero count, zero sum, Integer.MAX_VALUE min, Integer.MIN_VALUE max and zero average.

Method Detail

accept

```
public void accept(int value)
```

Records a new value into the summary information

Specified by:

`accept` in interface `IntConsumer`

Parameters:

`value - the input value`

combine

```
public void combine(IntSummaryStatistics other)
```

Combines the state of another `IntSummaryStatistics` into this one.

Parameters:

`other - another IntSummaryStatistics`

Throws:

`NullPointerException - if other is null`

getCount

```
public final long getCount()
```

Returns the count of values recorded.

Returns:

`the count of values`

getSum

```
public final long getSum()
```

Returns the sum of values recorded, or zero if no values have been recorded.

Returns:

`the sum of values, or zero if none`

getMin

```
public final int getMin()
```

Returns the minimum value recorded, or `Integer.MAX_VALUE` if no values have been recorded.

Returns:

`the minimum value, or Integer.MAX_VALUE if none`

getMax

```
public final int getMax()
```

Returns the maximum value recorded, or `Integer.MIN_VALUE` if no values have been recorded.

Returns:

`the maximum value, or Integer.MIN_VALUE if none`

getAverage

```
public final double getAverage()
```

Returns the arithmetic mean of values recorded, or zero if no values have been recorded.

Returns:

`the arithmetic mean of values, or zero if none`

toString

```
public String toString()
```

Description copied from class: `Object`

Returns a string representation of the object. In general, the `toString` method returns a string that "textually represents" this object. The result should be a concise but informative representation that is easy for a person to read. It is recommended that all subclasses override this method.

The `toString` method for class `Object` returns a string consisting of the name of the class of which the object is an instance, the at-sign character `` `@' ``, and the unsigned hexadecimal representation of the hash code of the object. In other words, this method returns a string equal to the value of:

```
getClass().getName() + '@' + Integer.toHexString(hashCode())
```

Overrides:

`toString` in class `Object`

Returns:

a string representation of the object.

C

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